

UNIT 4

ROTATIONAL AND CIRCULAR MOTION



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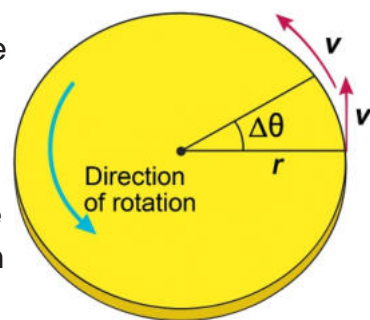


ANGULAR DISPLACEMENT:

"The angle swept by the position vector of a body moving in a circle relative to the centre of the circle is called angular displacement."

Measurement of angular displacement:

Consider a particle is moving in circle. It moves from point P1 to P2 such that its position vector 'r' which represents the radius of circle making an angle 'θ' from origin. Whereas 'S' is the length of arc from P1 to P2. The angular displacement is given by,



$$\text{Angular displacement} = \frac{\text{Length of arc}}{\text{Radius of circle}}$$

$$\theta = \frac{S}{r}$$

$$S = r \theta$$

Representation:

Angular displacement is denoted by 'θ'.

Nature of quantity:

For very small value of Δθ, the angular displacement is vector quantity.

Direction:

The direction associated with Δθ is along the axis of rotation.

Units:

The angular displacement is measured in degree or radian.

DEFINITIONS OF UNITS OF ANGULAR DISPLACEMENT:**• Degree:**

"If one complete revolution divided into 360 equal parts then the angular displacement of single part is called one degree."

• Radian:

"If the angle subtended at the centre of a circle by an arc equal to the length of its radius then angular displacement is said to be one radian."

Relation between degree and Radian:

When particle completes one revolution its angular displacement is degree is 360°.

$$\theta = 360^\circ \dots\dots\dots (i)$$

We know that the circumference of circle is 2πr. It the total arc length of the circle.

$$S = 2\pi r$$

Angular displacement in radian is,

$$\theta = \frac{S}{r}$$

$$\theta = \frac{2\pi r}{r}$$

$$\theta = 2\pi \dots\dots\dots (ii)$$

By comparing equation (i) and (ii),

$$2\pi \text{ radian} = 360 \text{ degree}$$

$$1 \text{ Radian} = \frac{360}{2\pi} = 57.3 \text{ Degree}$$

$$1 \text{ Degree} = \frac{2\pi}{360} = 0.01745 \text{ Radian}$$

ANGULAR VELOCITY:

"Angular displacement per unit time is called angular velocity."

Representation:

It is denoted by ' ω '.

Formula:

If a rotating body covers angular displacement $\Delta\theta$ in time Δt , then its angular velocity will be,

$$\text{Angular velocity} = \frac{\text{Angular displacement}}{\text{Time taken}}$$

$$\omega = \frac{\Delta\theta}{\Delta t}$$

Nature:

Angular velocity is a vector quantity.

Direction:

Angular velocity is directed along the axis of rotation and can be found out by using right hand rule.

Units:

- Radian per second (rad/sec).
- Revolution per minutes (rpm) [1 rpm = $2\pi / 60$ Rad/Sec]
- Degree per second (deg/sec).
- Revolution per second (rev/sec).

INSTANTANEOUS ANGULAR VELOCITY:

"Angular velocity of a particle at a particular instant of time is called instantaneous angular velocity."

Formula:

$$\omega_{\text{inst}} = \lim(\Delta t \rightarrow 0) \frac{\Delta\theta}{\Delta t}$$

Relation between linear and angular velocity:

Consider a particle rotating in a circular path of radius ' r '. Let the particle rotates through an angle $\Delta\theta$ in time Δt making an arc of length ΔS . Then by using relation $S = r\theta$, it can be written as,

$$\frac{\Delta S}{r} = \Delta\theta$$

$$\Delta S = r\Delta\theta$$

Dividing both sides by Δt we get,

$$\frac{\Delta S}{\Delta t} = r \times \frac{\Delta\theta}{\Delta t}$$

If time interval Δt is very small ($\Delta t \rightarrow 0$), then,

$$\lim(\Delta t \rightarrow 0) \frac{\Delta S}{\Delta t} = r \lim(\Delta t \rightarrow 0) \frac{\Delta\theta}{\Delta t}$$

Where,

$$\lim(\Delta t \rightarrow 0) \frac{\Delta S}{\Delta t} = V_T \text{ and } \text{formula} = \lim(\Delta t \rightarrow 0) \frac{\Delta\theta}{\Delta t} = \omega$$

Therefore,

$$V_T = r\omega$$

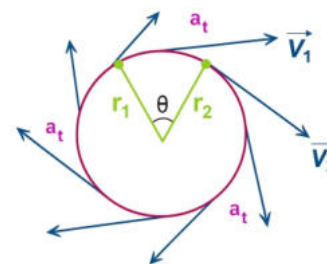
TANGENTIAL VELOCITY:

"Velocity of a rotating particle that is tangent to the circular path is called tangential velocity."

Expression for tangential velocity:

$$V_T = r\omega$$

Relation $V_T = \omega r$ is valid only when the angular speed ' ω ' of the rotating body is measured in radian per unit time.



ANGULAR ACCELERATION:

"Change in angular velocity per unit time is called angular acceleration."

Representation:

It is denoted by ' α '.

Formula:

If a rotating body covers angular velocity $\Delta\omega$ in time Δt , then its angular acceleration will be,

$$\text{Angular acceleration} = \frac{\text{Change in angular velocity}}{\text{Time taken}}$$

$$\alpha = \frac{\Delta\omega}{\Delta t}$$

Nature:

Angular acceleration is a vector quantity.

Direction:

Angular velocity is directed along the axis of rotation and can be found out by using right hand rule.

Units:

- Radian/second square (rad/sec^2).
- Degree/second square (deg/sec^2).
- Revolution/second square (rev/sec^2).

INSTANTANEOUS ANGULAR ACCELERATION:

"Angular acceleration of a particle at a particular instant of time is called instantaneous angular acceleration."

Formula:

$$\alpha_{\text{inst}} = \lim(\Delta t \rightarrow 0) \frac{\Delta\omega}{\Delta t}$$

Relation between tangential and angular acceleration:

The linear or tangential velocity of rotating body is given by,

$$V_T = r\omega$$

Suppose an object rotating about a fixed axis, changes its angular velocity by $\Delta\omega$ in time ' t '. Then the change in tangential velocity ΔV_T at the end of this interval is,

$$\Delta V_T = r\Delta\omega$$

Dividing both sides by Δt , we get,

$$\frac{\Delta V_T}{\Delta t} = r \frac{\Delta\omega}{\Delta t}$$

If time interval Δt is very small ($\Delta t \rightarrow 0$), then,

$$\lim(\Delta t \rightarrow 0) \frac{\Delta v_t}{\Delta t} = r \lim(\Delta t \rightarrow 0) \frac{\Delta \omega}{\Delta t}$$

Where,

$$\lim(\Delta t \rightarrow 0) \frac{\Delta v_t}{\Delta t} = a_T \text{ and } \lim(\Delta t \rightarrow 0) \frac{\Delta \omega}{\Delta t} = \alpha$$

Therefore,

$$a_T = r\alpha$$

TIME PERIOD:

"Time taken by a body to complete one revolution is known as its time period or period of revolution."

Representation:

It is denoted by 'T'.

Expression for time period of rotating body:

According to the definition of angular velocity we can write as,

$$\omega = \frac{\theta}{t}$$

For one complete revolution $\theta = 2\pi$ radians. Therefore,

$$\omega = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{\omega}$$

CENTRIPETAL ACCELERATION OR RADIAL ACCELERATION:

"The acceleration produced by virtue of changing direction of the velocity of an object moving in circular path is called centripetal acceleration."

Representation:

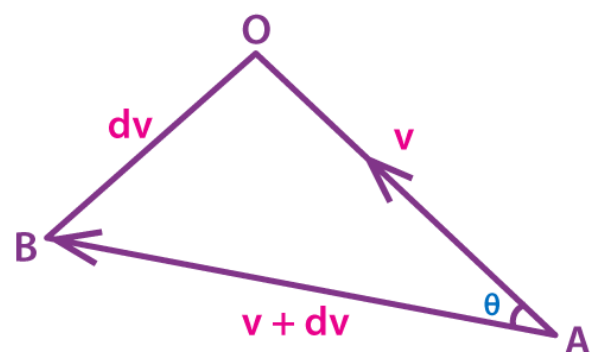
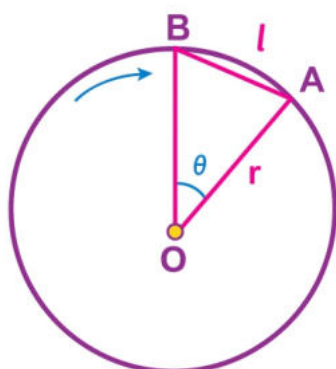
It is denoted by a_c .

Direction:

Centripetal acceleration is directed towards the center of the circle.

Expression for centripetal acceleration:

Consider a particle moving in a circle of radius 'r' centered at 'O' with constant speed 'V'. It goes from point P_1 to P_2 in time Δt along arc length ΔS . Angular displacement during this time is ' θ '. Let V_1 and V_2 denote the instantaneous linear velocities at point P_1 and P_2 respectively. Change of velocity vector is given by:



$$\Delta V = V_2 - V_1$$

V_1 and V_2 are perpendicular to the radius of circle at points A and B. Therefore, the angle between V_1 and V_2 is also ' θ '. ΔP_1OP_2 and ΔBAC are isosceles triangle and are similar, therefore,

$$\frac{\Delta V}{V} = \frac{\Delta s}{r}$$

$$\Delta V = \frac{v}{r} \Delta s$$

Dividing both sides by Δt we get,

$$\frac{\Delta V}{\Delta t} = \frac{v}{r} \cdot \frac{\Delta s}{\Delta t}$$

Applying limit as Δt approaches to zero,

$$\lim(\Delta t \rightarrow 0) \frac{\Delta V}{\Delta t} = \frac{v}{r} \lim(\Delta t \rightarrow 0) \frac{\Delta s}{\Delta t}$$

We know that,

$\lim(\Delta t \rightarrow 0) \frac{\Delta V}{\Delta t} = a_c$ & $\lim(\Delta t \rightarrow 0) \frac{\Delta s}{\Delta t} = v$ by substituting these values in above equation,

$$a_c = \frac{v}{r} \cdot v$$

$$a_c = \frac{v^2}{r}$$

$$a_c = \frac{v^2}{r}$$

We know that $v = r\omega$,

$$a_c = \frac{(r\omega)^2}{r}$$

$$a_c = \frac{r^2\omega^2}{r}$$

$$a_c = r\omega^2$$

Where $\omega = \frac{2\pi}{T}$

$$a_c = r\left(\frac{2\pi}{T}\right)^2$$

$$a_c = \frac{4\pi^2 r}{T^2}$$

TANGENTIAL ACCELERATION:

"The acceleration produced in a body during circular motion tangent to the curve of circle is called tangential acceleration"

Representation:

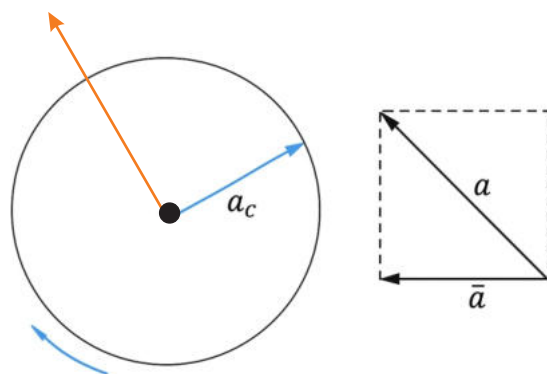
It is represented by a_T .

Direction:

It is always directed tangent to the curve of circle.

RELATION BETWEEN TANGENTIAL AND CENTRIPETAL ACCELERATION:

Consider a body moving in a circular path, in which both components of acceleration i.e. tangential and centripetal acceleration exist simultaneously.



Resultant acceleration:

These two components of acceleration are perpendicular to each other than total acceleration a by using vector diagram is given by,

$$a = a_c + a_T$$

Magnitude of resultant acceleration:

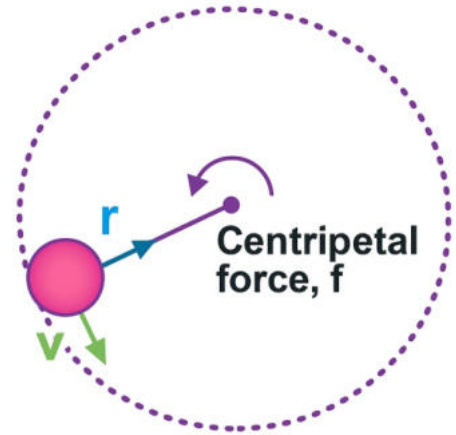
The magnitude of total acceleration a is given by,

$$a = (\sqrt{a_c^2 + a_T^2})$$

Direction of resultant acceleration:

The direction of a with respect to a_c is,

$$\theta = \tan^{-1}\left(\frac{a_T}{a_c}\right)$$

**CENTRIPETAL FORCE:**

"when a body moves in a circle with uniform speed the force required to keep it moving in a circular path is called centripetal force."

Mathematical representation:

If a body of mass ' m ' is moving in a circle of radius ' r ' with a velocity ' v ' then according to the definition of centripetal acceleration,

$$a_c = \frac{v^2}{r}$$

According to Newton's 2nd law of motion,

$$F_c = m a_c$$

Where $v = r\omega$,

$$F_c = m \frac{(r\omega)^2}{r}$$

$$F_c = \frac{mr^2\omega^2}{r}$$

$$F_c = mr\omega^2$$

$$\text{Where } \omega = \frac{2\pi}{T}$$

$$F_c = m r \left(\frac{2\pi}{T}\right)^2$$

$$F_c = \frac{4\pi^2 mr}{T^2}$$

Direction:

Centripetal force is always directed towards the center of circle.

FACTORS AFFECTING CENTRIPETAL FORCE:**• Mass of the body:**

Centripetal force is directly proportional to the mass of the body.

• Velocity of the body:

Centripetal force is directly proportional to the square of the velocity of the body.

• Radius of circle:

Centripetal force is inversely proportional to the radius of circle.

CENTRIFUGAL FORCE:

"The force which is directed away from the center of the circle is known as centrifugal force."

CENTRIPETAL ACCELERATION CAUSED BY TENSION FORCE OR VERTICAL CIRCULAR MOTION:

The centripetal force (F_c) keeps the object in its circular path. It is expressed as

$$F_c = \frac{mv^2}{r}$$

In an object attached to a string while moving in a vertical circular motion, the centripetal force is the sum of the object's weight (mg) and the tension in the string (T).

The value of the tension varies in the different parts of the vertical circle and has different vertical circular motion equations associated with it.

Tension at the bottom of the circle:

Tension is at its maximum at the bottom of the circle (T_B), and its direction is opposite that of the object's weight.

In this case, the centripetal tension formula would be:

$$T_B - mg = \frac{mv^2}{r}$$

$$T_B = \frac{mv^2}{r} + mg$$

Tension at the top of the circle:

On the other hand, the tension at the top of the circle (T_T) is at its minimum and is in the same direction as the object's weight.

$$T_t + mg = \frac{mv^2}{r}$$

$$T_t = \frac{mv^2}{r} - mg$$

BANKED CURVE:**Definition:**

"Banking of roads is defined as the phenomenon in which the outer edges are raised for the curved roads above the inner edge to provide the necessary centripetal force to the vehicles so that they take a safe turn."

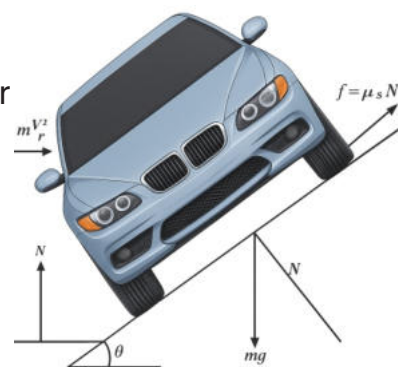
Expression for angle of banking:

Let us now consider banked curves, where the slope of the road helps you negotiate the curve. The greater the angle θ , the faster you can take the curve.

If the angle θ is ideal for the speed and radius, then the net external force equals the necessary centripetal force. The only two external forces acting on the car are its weight W and the normal force of the road N .

Only the normal force has a horizontal component, so this must equal the centripetal force, that is,

$$N \sin \theta = \frac{mv^2}{r} \text{ ----- (1)}$$



From (Figure), we see that the vertical component of the normal force is $N \cos \theta$, and the only other vertical force is the car's weight. These must be equal in magnitude; thus,

$$N \cos \theta = mg \text{ ----- (2)}$$

Dividing equations (1) and (2):

$$\frac{N \sin \theta}{N \cos \theta} = \frac{mv^2}{r} \div mg$$

$$\frac{\sin \theta}{\cos \theta} = \frac{mv^2}{r} \frac{1}{g}$$

$$\tan \theta = \frac{v^2}{rg}$$

$$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$$

ORBITAL VELOCITY:

"The velocity with which the satellite moves around the earth is called orbital velocity."

Derivation:

Consider a satellite of mass "m" moving in an orbit of radius "R = (R_e + h)" with velocity "V" around the earth. The gravitational force of attraction between the satellite and the earth provides necessary centripetal force.

If M is the mass of the earth, then

$$\text{Gravitational force} = \frac{GMm}{R^2} \text{ \& Centripetal force} = \frac{mv^2}{r}$$

Where $F_c = F_g$

$$\frac{GM_e m}{R^2} = \frac{mv^2}{R}$$

Or

$$v^2 = \frac{GM_e}{R} \text{ ----- (i)}$$

By taking square root on both sides,

$$v = \sqrt{\left(\frac{GM_e}{R}\right)}$$

If satellite revolving near to the surface of the earth, then $R = R_e$ = radius of earth and the value of orbital velocity becomes,

[From astronomical data, $g = 9.8 \text{ m/s}^2$ and $R_e = 6.4 \times 10^6 \text{ m}$]

$$v = \sqrt{(9.8 \text{ m/s}^2)(6.4 \times 10^6 \text{ m})}$$

$$v = 7920 \text{ m/s OR } v = 8 \text{ Km/s (Approx.)}$$

TIME PERIOD OF ARTIFICIAL SATELLITE:

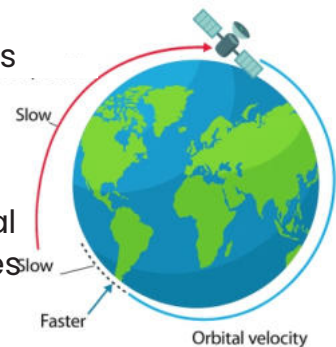
Definition:

"The time required for a satellite to complete one revolution around the Earth in its orbit is called its time period "T"."

Expression for time period:

The time period of a satellite can be calculated as:

$$T = \frac{2\pi R}{v}$$



Where $R = R_e + h$, By substituting the value of 'v' in equation of orbital velocity,

$$T = 2\pi R \sqrt{\frac{R}{GM_e}}$$

$$T = 2\pi \sqrt{R^2} \sqrt{\frac{R}{GM_e}}$$

$$T = 2\pi \sqrt{\frac{R \times R^2}{GM_e}}$$

$$T = 2\pi \sqrt{\frac{R^3}{GM_e}}$$

MOMENT OF INERTIA OR ROTATIONAL INERTIA OR ANGULAR MASS

"The moment of inertia is defined as the quantity expressed by the body resisting angular acceleration, which is the sum of the product of the mass of every particle with its square of the distance from the axis of rotation."

Moment of inertia = Total mass x Square of distance from center

$$I = mr^2$$

Unit & Dimensions:

The SI unit of moment of inertia is kg m^2 and dimension is $M^1 L^2 T^0$.

Factors of moment of inertia:

- The density of the material
- Shape and size of the body
- Axis of rotation (distribution of mass relative to the axis)

Moment of inertia of two particle system:

Consider a rigid body containing of two particles of masses m connected by a rod of length L with negligible mass.

Case – I

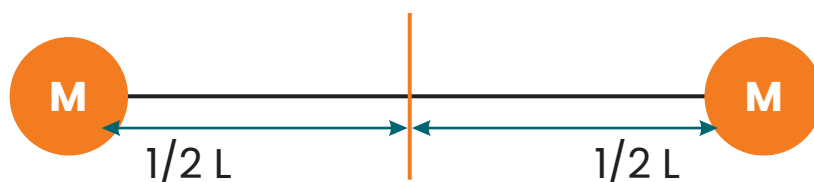
Two particles each at perpendicular distance $\frac{1}{2} L$ from axis of rotation. For two particles each at perpendicular distance $\frac{1}{2} L$ from the axis of rotation we have,

$$I = \sum m r^2$$

$$I = m \left(\frac{1}{2} L\right)^2 + m \left(\frac{1}{2} L\right)^2$$

$$I = m \left(\frac{1}{4} L^2\right) + m \left(\frac{1}{4} L^2\right)$$

$$I = m \left(\frac{1}{2} L^2\right)$$



Case – II

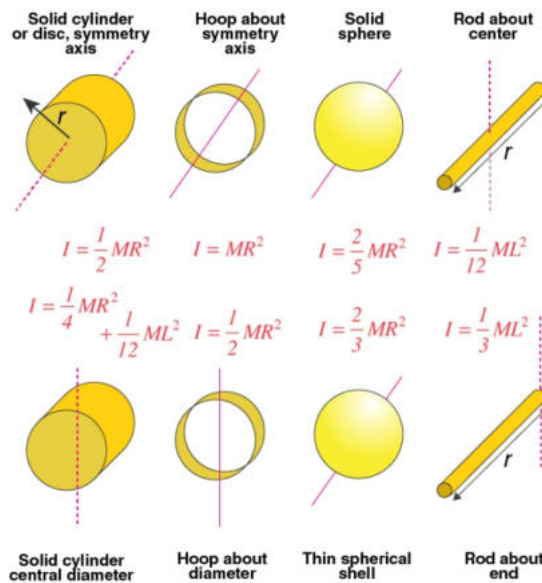
For two particles the axis of rotation is one end of any mass 'm' we have,

$$I = \sum m r^2$$

$$I = m (0)^2 + m (L)^2$$

$$I = mL^2$$

Moment of inertia of various shapes:



ANGULAR MOMENTUM:

"Angular momentum is defined as the vector product of linear momentum and position vector."

Mathematical expression:

Consider a particle of mass 'm' let its position vector $\rightarrow r$ and its linear momentum $\rightarrow P$ measured with origin 'O'.

Since by the definition of angular momentum,

$$L = r \times P$$

$$L = rP \sin \theta$$

If a body is moving in a circle, then angle between v and r is 90° .

$$L = mvr \sin \theta$$

If the body is moving in a circle, then angle between v and r is 90° .

$$L = mvr$$

Dimension:

The dimension of angular momentum is.

$$ML^2T^{-1}$$

Units:

The units of angular momentum are:

$$Kg.m^2/s \text{ or } Kg \text{ m}^2s^{-2}$$

Multiplying and dividing above by 's'

$$J\text{-Sec}$$

LAW OF CONSERVATION OF ANGULAR MOMENTUM:

Statement:

"If no net external torque acts on a body or system, its angular momentum remains constant in both magnitude and direction."

Proof:

According to Newton's 2nd law of motion net force acting on a particle of mass 'm' moving with an instantaneous velocity $\vec{V}$ is the time rate of change of its linear momentum. Thus F stand $\vec{F}$ for force and $\vec{P}$ for linear momentum,

By the definition of angular momentum,

$$\vec{L} = \vec{r} \times \vec{p}$$

Differentiate above equation with respect to time,

$$\frac{d}{dt}(L) = \frac{d}{dt}(r \times p)$$

$$\frac{d}{dt}(L) = \vec{r} \times \frac{d}{dt}(p) + \frac{d}{dt}(r \times p)$$

$$\frac{d}{dt}(L) = \vec{r} \times F + v \times p$$

$$\frac{d}{dt}(L) = \vec{r} \times F + m(v \times v)$$

$$\frac{d}{dt}(L) = \vec{r} \times F + v \times mv$$

$$\frac{d}{dt}(L) = \vec{r} \times F + m \times 0$$

Formula

$$\frac{d}{dt}(L) = \vec{r} \times F$$

$$\frac{d}{dt}(r) = v \text{ \& } \frac{d}{dt}(p) = F$$

$$r \times F \text{ \& } p = mv$$

$$v \times v = 0$$

$$\frac{d}{dt}(L) = \tau$$

Above relation shows that rate of change of angular momentum gives the value of torque
If the net torque remains zero, then,

$$\frac{d}{dt}(L) = 0$$

$$L = \text{Constant}$$

RELATION BETWEEN ANGULAR MOMENTUM AND MOMENT OF INERTIA:

From the definition of angular momentum,

$$L = mvr$$

Where for angular motion $v = r\omega$,

$$L = m(r\omega)r$$

$$L = (mr^2)\omega$$

From the definition of moment of inertia $mr^2 = I$,

$$L = I\omega$$

TORQUE:

“The turning effect produced in a body about an axis of rotation is called torque.”

Mathematically:

Let F be the force and r be the position vector, mathematically torque is written as,

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Magnitude of torque:

$$|\vec{\tau}| = rF\sin\theta$$

Where θ is the angle between $\vec{r}$ & $\vec{F}$.

Direction of torque:

Torque vector $\vec{\tau}$ is always directed along the normal to the plane defined by $\vec{r}$ & $\vec{F}$.

Unit of torque:

The S.I. unit of torque is Nm.

Dimension of torque:

Dimension of torque is ML^2T^{-2} .

RELATION BETWEEN TORQUE AND MOMENT OF INERTIA:

From the definition of torque,

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = r \times ma$$

$$\tau = r \times m(r\alpha)$$

$$\tau = m r^2 \alpha$$

$$\tau = I \alpha$$

LINEAR MOTION WITH CONSTANT ACCELERATION ' α '	ANGULAR MOTION WITH CONSTANT ACCELERATION ' α '
$S = Vt$	$\theta = \omega t$
$V_f = V_i + at$	$\omega_f = \omega_i + at$
$V_{av} = (V_f + V_i) / 2$	$\omega_{av} = (\omega_f + \omega_i) / 2$
$S = V_i t + (1/2) at^2$	$\theta = \omega_i t + (1/2) at^2$
$2as = V_f^2 - V_i^2$	$2\alpha s = \omega_f^2 - \omega_i^2$
$P = mV$	$L = mvr$